

COURSE:	Programming in C++ CS 190 Section 70
PRE/CO-REQUISITES	CS 110 and Eligibility for college level English based on the SBVC Guided-Self Placement process.
MEETS:	This is a fully online class. Students will be required to complete online instructional content via Canvas. This class could offer some live online meetings, but students will have options on accessing that content. Students should log on to Canvas the first day of class to access classes.
PROFESSOR:	Cristian I. Racataian
E-MAIL	cracatai@sbccd.cc.ca.us
VIRTUAL OFFICE HOURS:	Mondays, 1:00 pm to 2:00 pm Join URL: https://sbccd-edu.zoom.us/j/82828010640 No appointment needed. Students can join in any time during the hour. As part of Regular Substantive Interaction (RSI) , I offer weekly office hours for all my students. These office hours are not mandatory; however, they offer a great resource for questions you might have about assignments, lecture materials, or your grade. It is helpful to send me a Canvas Message to let me know what you'd like to discuss.
PHONE:	See division info.
DESCRIPTION:	This course is an examination of intermediate object-oriented programming concepts and their application using the C++ language. Topics include programming control mechanisms; algorithm development; analysis of iterative and recursive solution complexity for various algorithms; templates and data structures; exception handling; object-oriented design and modeling; object-oriented programming in software engineering; pointers architecture.
TEXT:	Starting Out with C++: Early Objects, 10 th edition, Tony Gaddis, Judy Walters, and Godfrey Muganda, 2020, Pearson Education, ISBN-10: 0135235006, ISBN-13: 978-0135235003
SOFTWARE:	Microsoft Visual Studio Community (free edition) Download web address: https://visualstudio.microsoft.com/ Visual Studio is recommended because is installed in the lab and is available for both PC and Mac OS. Any other ANSI compliant IDE can be used to develop C++ programs for this course.
STUDENT LEARNING OUTCOMES	Upon successful completion of the course, students should be

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able to:

1. Implement and analyze the coding of various algorithms.
2. Construct multiple files or multiple modules programming solutions that use class hierarchies, inheritance, and polymorphism to reuse existing design and code.
3. Construct object-oriented programming solutions for reuse, using ADTs and templates that incorporate encapsulation, data abstraction, and information hiding.
4. Create programming solutions that use data structures and existing libraries.
5. Design and develop algorithms utilizing iterative repetitions and recursive repetitions.

OBJECTIVES:

1. Implement and analyze the coding of various algorithms.
2. Construct multiple-file or multiple-module programming solutions that use class hierarchies, inheritance, and polymorphism to reuse existing design and code.
3. Construct object-oriented programming solutions for reuse, using abstract data types and templates that incorporate encapsulation, data abstraction, and information hiding.
4. Create programming solutions that use data structures and existing libraries.
5. Design and develop algorithms utilizing iterative repetitions and recursive repetitions; analyze the advantages vs. the pitfalls.
6. Discuss significant trends and societal impacts of computing, software, and the Internet.
7. Exam the architecture of pointers, the pros and cons of using the pointer mechanism.
8. Ensure program correctness through the implementation of exception throwing, catching, and handling.

COURSE WEB PAGE:

Canvas

**INSTRUCTOR
REQUIREMENTS:**

Assignments 40% -- five assignments worth 80 pts. each (400 pts)
Lab work: 30% -- twelve assignments worth 25 pts. each (300 pts)
Midterm exam: 15% -- during the 9th week (150 pts)
Final exam: 15% -- on the final week (non-cumulative, 150 pts)

GRADING:

900 – 1000 A ; 800 – 899 B ; 700 – 799 C ; 560 – 699 D; 0 – 559 F

**ONLINE COURSE
REQUIREMENTS**

Students are expected every week to login to Canvas and work with the current module for chapter reading and completing the chapter lab work. For clarifications and/or course related communications, students should use Canvas email and/or Message Board. Specific instructions

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INTELLIGENCE (AI)
USAGE:**

will be posted on Canvas.

Students are **NOT** permitted to use AI agents for class work even if the final product complies exactly with the assignment requirements. Submissions that produce similar results but do not comply with the requirements will not be accepted.

Please consider that this course is about fundamental concepts in C++ programming. AI is in its infancy, and the time it takes to generate the exact same code implementation would be much better spent in acquiring the more valuable programming skill.

ATTENDANCE:

All students are expected to login onto Canvas regularly. In the event a period of absence is unavoidable, students are responsible for notifying instructors. Failure to login during the first week of the class may result in the student being dropped from the class.

After the first week of class, a student may be dropped if:

- The student fails to login regularly without an excused absence;
- The student has more than two weeks without login;
- The student login is so irregularly that makes it unlikely to succeed.

CHEATING POLICY:

Cheating constitutes academic dishonesty and, in general will be handled as part of the course grading process. Penalty may range from no credit for the assignment up to and including exclusion and/or an "F" grade for the course.

ADA Accommodations**Accommodations for a Student with a Disability**

In compliance with the Americans with Disabilities Act (ADA), all qualified students enrolled in this course are entitled to "reasonable accommodations." Please notify the instructor during the first week of class of any accommodations needed for the course.

If you have a disability (e.g., physical, learning, etc...) documented by a physician or another appropriate licensed professional, please be sure to contact Student Accessibility Services (SAS) to discuss appropriate academic accommodations. Additionally, you may but are not required to contact me directly to ensure that proper accommodations, as deemed by SAS, are acknowledged and carried out most effectively and appropriately. For further information, contact SAS at (909) 384-4443 or visit the SAS office in the Administration/Student Services Building, Room 105, to make an appointment for intake and advisement.

This syllabus/outline may be subject to additions, deletions, and/or changes at the instructor's discretion or San Bernardino Valley College/SBCCD. The course content may vary from this



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outline to meet the needs of this particular group. Advance notice of such change shall be given.

THE LIBRARY

Open Monday – Saturday. Call 384-8841 for information and current hours.

THE ACADEMIC ADVANCEMENT LAB

Provides access to computers, software, printers, and tutors. Call 384-4447 for current hours and additional information.

COUNSELORS

Available to assist students with career and curriculum planning decisions, as well as personal problems such as drug and alcohol dependency. Call 384-4404 for current hours or additional information.

Help Desk

Available for 24/7 Assistance with technical matters (e-mail, Canvas, Registration, Campus Central Log-in, etc.) **(877) 241-1756**

DIVISION:

Academic Services, Business, Computer Technology & Hospitality

DIVISION OFFICE:

San Bernardino Valley College, B 127

DIVISION SECRETARY:

Vivian Marquez, B 127, 909 384 8520

IMPORTANT DATES:

Last day to add: September 4, 2026

Last day to drop without a grade: September 13, 2026

Last day to withdraw with a W: November 6, 2026

Final Exam: December 12 --18 online.

WEEKLY CALENDAR Dates may be adjusted as necessary. Students will be notified of all changes.	TOPICS Canvas Module/Textbook Chapter	LAB WORK	ASSIGNMENT Dates**	EXAMS
Week 1: 8/24-30	Introduction. Canvas Content.			
Week 2: 8/31-9/6	Mod. 1/ Chap. 1			
Week 3: 9/7-13	Mod. 1/ Chap. 2	Lab Ch.2		
Week 4: 9/14-20	Mod. 1/ Chap. 3	Lab Ch.3	Homework #1	
Week 5: 9/21-27	Mod. 2/ Chap. 4	Lab Ch.4		
Week 6: 9/28-10/4	Mod. 2/ Chap. 5	Lab Ch.5		
Week 7: 10/5-11	Mod. 2/ Chap. 6	Lab Ch.6	Homework #2	
Week 8: 10/12-18	Mod. 3/ Chap. 7	Lab Ch.7		
Week 9: 10/19-25	Mod. 3/ Chap. 8	Lab Ch.8		Midterm Chapters 1-6
Week 10: 10/26-11/1	Mod. 3/ Chap. 9	Lab Ch.9	Homework #3	
Week 11: 11/2-8	Mod. 4/ Chap. 10	Lab Ch.10		
Week 12: 11/9-15	Mod. 4/ Chap. 11	Lab Ch.11		
Week 13: 11/16-22	Mod. 4/ Chap. 12	Lab Ch.12	Homework #4	
Fall Break: 11/23-29	Thanksgiving Recess			
Week 14: 11/30-12/6	Mod. 5/ Chap. 13*			
Week 15: 12/7-13	Mod. 5/ Chap. 14	Lab Ch.14		
Week 16: 12/14-18	Mod. 5/ Chap. 15*		Homework #5	Final: Chapters 7-12,14

* Chapter will have a partial coverage and will NOT be included in the Final Exam.

** Each assignment is due within one week after the end of a module.